BIMS-LEGAL: Dual-Store Soft Binding for Recovering Prior Legal Advice under Same-Domain Interference

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Abstract

In same-domain legal consultation archives, dense retrieval often surfaces a relevant client question while failing to return the lawyer's advice turn—an *episode incompleteness* failure that leaves prior recommendations hard to recover under lexical interference from neighbouring matters. BIMS-LEGAL indexes Chinese consultation logs in a dual store: a fast episodic turn index with session identifiers and a slower semantic store consolidated by target-k KMeans. Soft O2 binds session siblings by attenuated score inheritance (βs) rather than hard score copy. Exact replay is a surface-form reference; main channels are paraphrase, follow-up, and advice-recall. On a shared turn-level FlatIP rebuild, Soft O2 raises Answer Hit over dense FlatIP; hard hydration, which copies sibling scores without attenuation, can raise AH further by forcing siblings into top-k. Soft O2-C is evaluated as a conditional cluster-binding mechanism under Mix reconstructions with off-session gold; gated Hybrid is the composed operator when thematic co-membership helps. We report Answer Hit, Episode Completeness, graded nDCG with fixed gold-session IDCG, a four-way failure taxonomy, and Holm-adjusted Soft O2 vs FlatIP tests.

Keywords: legal information retrieval, consultation memory, soft episode binding, dual-store memory, cluster binding, Chinese legal corpora

1. Introduction

A junior associate may ask a firm assistant: "Last month you advised this client on terminating a fixed-term contract—what remedy did we recommend?" The system returns several near-identical "contract termination" questions, ranks the stored user utterance highly, and never surfaces the partner's written advice. The subsequent reply may sound fluent yet be grounded in the wrong episode—or in none. In consultation archives, questions and answers share specialized terminology, neighbouring matters are topically adjacent yet legally distinct, and the professionally useful evidence unit is the full consultation *episode*.

Legal informatics has long studied statutory interpretation, case retrieval, and advisory systems Martinez-Gil (2023); Ashley (2017), and recent legal large language models (LLMs) further improve statutory and fact-pattern answering Yue et al. (2023); Huang et al. (2023); Yue et al. (2024). Classical legal information retrieval (IR), however, mainly targets *external* authority—statutes, cases, or community answers—as documents Askari et al. (2024); Louis et al. (2024); Liu et al. (2022); Zhang et al. (2024). Memory-augmented dialogue systems recover evidence from evolving conversation logs Wu et al. (2024); Maharana et al. (2024); Packer et al. (2023), yet firm assistants still need a reliable index of *internal consultation memory*: prior client-specific advice buried in a growing same-domain archive under lexical interference from neighbouring matters.

Under that interference, dense turn retrieval often returns a question turn while its answer sibling remains outside top-k. We term this failure *episode incompleteness*, as distinct from a full session miss. It is worsened when approximate indexes distort turn scores and fragment episode context at moderate archive sizes. Parent-document hydration and session-max aggregation can force siblings into the result list by hard score copy, but they also erase rank competition among unrelated candidates. What is needed is a way to complete episodes without collapsing the retrieval ranking into a session dump.

This paper presents BIMS-LEGAL, a dual-store system for Chinese legal consultation logs. A fast episodic pathway indexes timestamped turn embeddings with speaker role and session identifier *sid*; a slower semantic pathway consolidates turns with target-k KMeans. The dual-pathway split is inspired by Complementary Learning Systems theory McClelland et al. (1995); Kumaran et al. (2016) only as an architectural analogy. On the episodic pathway, Soft O2 performs session soft binding: when a turn is retrieved with score s, siblings sharing *sid* inherit $\max(s_{t'},\ \beta \cdot s)$, so an answer turn can surface under question-hit / answer-miss while attenuated scores keep candidate competition intact. On the semantic pathway, Soft O2-C applies the same rule within KMeans clusters when gold evidence is not co-session with the query; gated Hybrid is intended to trigger cluster binding only on direct dense hits. Exact FlatIP indexing (O1) restores faithful turn scores when product quantization collapses Answer Hit; bulk-load consolidation (O3) makes large shared-store construction tractable without changing retrieval semantics. We evaluate Soft O2 primarily against FlatIP under episode incompleteness, treat hard hydration as an unrestricted score-copy upper bound, and examine Soft O2-C under Mix reconstructions as a conditional mechanism test.

The main contributions are as follows. First, we formulate consultation-memory retrieval under same-domain interference and identify episode incompleteness as a recurrent failure mode of dense turn retrieval. Second, we propose Soft O2 as a turn-level soft-binding operator: a gap-ratio score-competition analysis identifies a feasible high-β band, and an Answer Hit grid search over $\{0.5, 0.7, 0.9, 0.95, 0.98, 1.0\}$ selects default β=0.98 (Table A.4), with O1/O3 as implementation operators for shared-store evaluation. Third, we evaluate Soft O2 on CAIL multi-turn and rebuilt LegalEp corpora against FlatIP, hard hydration, shuffled-*sid*, BM25-turn, RRF, and cross-encoder controls under a unified FlatIP-rebuild metric pipeline, and report Soft O2-C Mix contrasts as conditional mechanism evidence with graded nDCG, a four-way failure taxonomy, and Holm-adjusted Soft O2 vs FlatIP tests.

Section 2 reviews related work. Section 3 presents BIMS-LEGAL and the Soft O2 attenuation analysis. Section 4 details corpora, metrics, and protocols. Section 5 reports results. Section 6 discusses implications and limitations, and Section 7 concludes.

2. Related Work

2.1. Legal IR and long-horizon dialogue memory

Legal QA and retrieval systems typically ground generation in statutes, cases, or published advice Louis et al. (2024); Askari et al. (2024); Martinez-Gil (2023); Ashley (2017). Conversational legal case retrieval and event-aware similar-case ranking further show how interaction context and structured legal knowledge shape IR effectiveness Liu et al. (2022); Zhang et al. (2024). Those pipelines optimize access to external authority. Consultation memory retrieval targets a different evidence source: *prior consultation episodes* inside an agent's own archive. Confusing two clients' histories can produce fluent yet professionally inappropriate responses even when statute retrieval is correct. Domain LLMs such as DISC-LawLLM, Lawyer LLaMA, and LawLLM improve statutory and advisory generation Yue et al. (2023); Huang et al. (2023); Yue et al. (2024), but they still need a reliable memory index when prior advice must be recovered from a growing same-domain log.

Legal consultation archives add a further difficulty: high lexical overlap among topically adjacent matters, with the professionally useful evidence unit being the full question–answer episode. This setting motivates *dual-store* memory architectures that keep fast episodic turn retrieval separate from slower semantic clustering, and session-aware soft binding operators that recover answer turns under same-domain interference without collapsing rank competition among unrelated candidates.

2.2. Dual-store memory and episodic retrieval

Dual-store designs distinguish fast episodic encoding from slower semantic integration McClelland et al. (1995); Kumaran et al. (2016); Tulving et al. (1972); Atkinson and Shiffrin (1971). Memory architectures for dialogue agents extend context via hierarchical paging, summary banks, graphs, and neuro-symbolic retrieval Packer et al. (2023); Zhong et al. (2024); Rasmussen et al. (2025); Gutiérrez et al. (2024); Modarressi et al. (2023); Wang et al. (2024); Lewis et al. (2020); Wu et al. (2025), but most treat retrieved units as flat passages rather than *multi-turn consultation episodes* whose answer turn may rank below an unrelated question turn sharing legal vocabulary. LongMemEval and LoCoMo evaluate long-horizon recall and QA over personal conversation logs Wu et al. (2024); Maharana et al. (2024), while broader memory-agent surveys emphasize persistence beyond the context window Wu et al. (2025); Tan et al. (2025). Recent conversational RAG work stresses reuse of historical search evidence across turns Mo et al. (2026). BIMS-LEGAL positions its dual-store contribution at this intersection: episodic turn embeddings with *sid* for same-session recovery (Soft O2), and KMeans semantic centroids with Soft O2-C for cross-session recovery when query and gold evidence occupy different sessions but share thematic structure. Under dense same-domain interference, a question turn can occupy top-k while its answer sibling is absent. Parent-document hydration and session-max aggregation recover that sibling by hard score copy; Soft O2 instead injects attenuated inherited scores so episode completion does not erase rank competition among unrelated candidates.

2.3. Session-aware retrieval

Parent-document hydration, session aggregation, and joint session indexing are standard session-aware mechanisms for attaching sibling context to a dense hit. Soft episode binding (Soft O2) inherits $\beta \cdot s$ for siblings and keeps ranking competition among candidates. Parent-document and hierarchical retrieval similarly attach child evidence to a parent unit Callan et al. (1994); Dai et al. (2019); Zaheer et al. (2017); Karpukhin et al. (2020); Soft O2 differs by using attenuated turn-level score propagation rather than hard hydration or atomic session documents. Lexical BM25 Robertson and Zaragoza (2009), dense passage retrieval Karpukhin et al. (2020), dense–sparse Reciprocal Rank Fusion Cormack et al. (2009), and cross-encoder reranking Nogueira et al. (2019); Chen et al. (2024) provide complementary controls outside session membership. Exact flat indexes and product-quantized ANN backends Johnson et al. (2019) further affect whether sibling binding is applied over faithful turn scores.

Contrast with joint-episode indexing.. An alternative way to keep episode context is to concatenate full question–answer pairs into monolithic episode documents (joint indexing). Joint indexing can prevent episode incompleteness by construction, but it changes the evidence unit from individual turns to merged documents. That coarser unit weakens fine-grained turn matching—for example, isolating a specific advice sentence—and raises index memory and update cost when new turns arrive in an online consultation log. Soft O2 instead keeps turn-level representations and candidate competition, recovering missing siblings on demand via attenuated score inheritance (βs) without replacing the underlying turn index. Table 1 summarizes the session-structure contrasts used in the main grids.

Table 1: Session-aware mechanisms compared in this work.

Mechanism	**Unit**	**Expansion**	**Type**	**Rank**
Parent hydration	Session	Insert siblings	Hard	No
Session-max	Session	Session max	Hard	Same*
Joint episode index	Session doc	Atomic	N/A	N/A
Soft O2	Turn+sid	Soft βs	Soft	Yes
Soft O2-C	Turn+cluster	Soft $\beta_c s$	Soft	Yes

*On the reported corpora, session-max rankings coincide with hard parent hydration under full-sibling expansion; main tables report a single hard-expansion baseline. Soft O2 denotes session soft binding; Soft O2-C denotes cluster soft binding on the KMeans semantic pathway.

In short, the dual-store split is the system design, and Soft O2 /Soft O2-C are the operators evaluated on Chinese legal consultation archives.

3. Method: BIMS-LEGAL

3.1. Architecture overview

Figure 1 summarizes the pipeline. BIMS-LEGAL maintains complementary episodic and semantic stores over the same consultation archive.

Definition 1 (Episodic and semantic elements)**.** *An* episodic turn *(episodic element) is $e_i = (\mathbf{v}_i, t_i, c_i, \phi_i, sid_i)$, where $\mathbf{v}_i \in \mathbb{R}^d$ is an embedding, t_i a timestamp, c_i conversational context (including speaker role), ϕ_i surface features, and sid_i the consultation session identifier. We reserve* session turns *for the weaker notion of turns that share only sid under a grouping key, without requiring role, timestamp, or other episodic metadata to participate in scoring. Soft O2 binds over session siblings $\mathrm{Sib}(t) = \{t' : sid_{t'} = sid_t\} \setminus \{t\}$, but the indexed objects themselves remain episodic turns. A semantic element is $s_j = (\mathbf{c}_j, C_j, \kappa_j, \tau_j)$ with centroid $\mathbf{c}_j$, member set C_j, label κ_j, and formation statistics τ_j. Write $\mathrm{Clus}(t) = \{t' : cid_{t'} = cid_t\} \setminus \{t\}$ for semantic-cluster siblings.*

Definition 2 (Dual-store retrieval map)**.** *Given query q with embedding $\mathbf{v}_q$, base episodic retrieval returns a scored candidate set $\mathcal{R}_0(q) = \{(t, s_t)\}$ under Eq.* (1)*. Soft O2 produces $\mathcal{R}_{\mathrm{O2}}(q) = \mathsf{SoftBind}(\mathcal{R}_0(q); \mathrm{Sib}, \beta)$ (Section 3.3). Soft O2-C produces $\mathcal{R}_{\mathrm{O2C}}(q) = \mathsf{SoftBind}(\mathcal{R}_0(q); \mathrm{Clus}, \beta_c)$ (Section 3.5). Gated Hybrid composes session binding then cross-session cluster binding (Algorithm 2). The returned list is the top-k of the bound set under the rewritten scores.*

3.2. Base scoring and implementation operators (O1, O3)

For query q with embedding $\mathbf{v}_q$, episodic base retrieval scores a memory turn m by

$$\mathrm{score}(m) = \alpha\, S(\mathbf{v}_q, \mathbf{v}_m) + (1 - \alpha)\, R(t_m), \tag{1}$$

where S is cosine similarity on ℓ_2-normalized embeddings and

$$R(t_m) = \max\left(0,\ 1 - \frac{\Delta_t}{30\ \mathrm{days}}\right), \tag{2}$$

with Δ_t the elapsed time between query time and the turn timestamp. Unless stated otherwise, α=0.7. Eq. (1) is the ranking score used throughout the legal grids; Soft O2 /Soft O2-C rewrite sibling scores after this fusion step and do not replace Eq. (1).

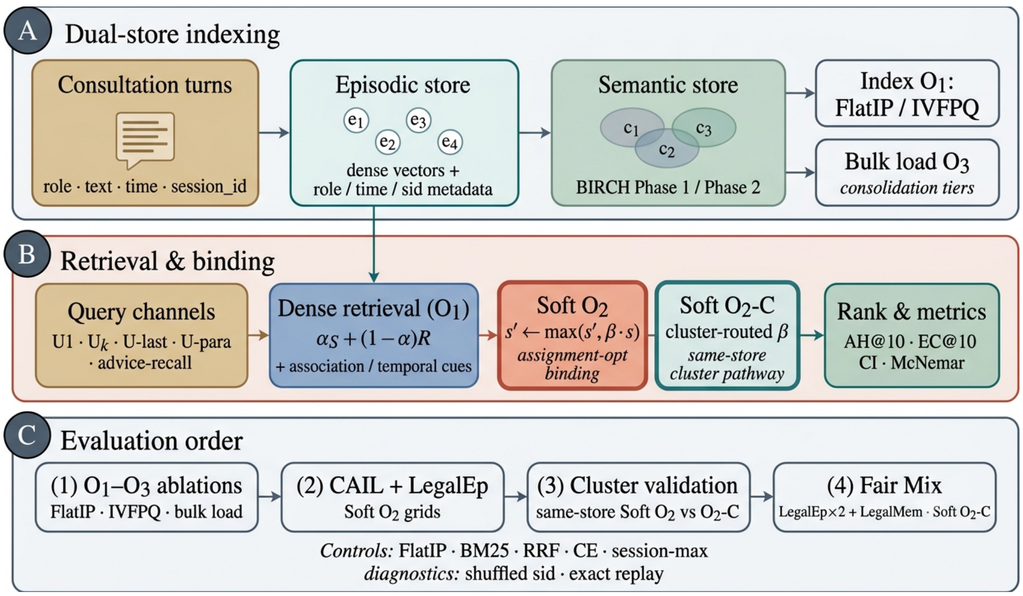

Figure 1: BIMS-LEGAL dual-store pipeline. (A) Episodic turn indexing and KMeans semantic consolidation (O3 bulk-load). (B) Query-time FlatIP retrieval with Soft O2 session binding and Soft O2-C cluster binding; O1 and O3 are implementation operators. (C) Evaluation order: shared-store ablations, Soft O2 grids, and Mix Soft O2-C.

O1: Exact FlatIP indexing.. Let $\widehat{S}$ denote the similarity induced by an ANN backend. At hundreds to a few thousand turns, FAISS IVFPQ is often undertrained, so $\|\widehat{S} - S\|$ is large enough to reorder turns and break Soft O2 inheritance. O1 forces $\widehat{S} = S$ via exact FlatIP (inner-product search on ℓ_2-normalized embeddings). Quantization error depresses Answer Hit (Table A.3: IVFPQ AH 0.330 at M=1600 on LegalEp-DISC exact). O1 is an engineering prerequisite for Soft O2 evaluation, not a standalone retrieval algorithm.

O3: Bulk-load consolidation.. During bulk construction, O3 replaces the per-insert update ($e_i \mapsto \mathsf{KMeansConsolidate}(e_i)$; Flush) by a deferred operator $\mathsf{Finalize} = \mathsf{KMeansConsolidate}(\{e_i\}_{i=1}^{N}) \circ \mathsf{Flush}$, executed once after all episodic writes. O3 does not change retrieval semantics; it makes $M \geq 400$ shared-store experiments runnable ($\approx$ 11 min wall-clock for M=400; amortized $\approx$ 0.825 s/turn over $\sim$ 800 turns). Quality–latency curves to M=6400 appear in Section 5.6.

Algorithm 1 Soft O2 session soft binding (implementation order)

Require: Query q, dense/associative hits $\mathcal{R}_0$, top-k, attenuation β

$\mathcal{R} \leftarrow \mathsf{ExpandSiblings}(\mathcal{R}_0; \beta)$ ▷ raw-score Soft O2
$\mathcal{R} \leftarrow \mathsf{ExactBoost/SparseAugment}(q, \mathcal{R})$ ▷ optional
$\mathcal{R} \leftarrow \mathsf{SessionFirstRerank}(\mathcal{R}, k)$
fuse each $r \in \mathcal{R}$ with Eq. (1)
$\mathcal{R} \leftarrow \mathsf{EnsureSessionCompleteness}(\mathcal{R}; \beta)$ ▷ final Soft O2 pass
return top-k of $\mathcal{R}$ by descending fused score

3.3. Soft O2: session soft binding

Motivation.. Under same-domain interference, a retrieved question turn can score highly while its answer sibling remains outside top-k—episode incompleteness compounded by IVFPQ score collapse and context fragmentation. Soft O2 completes episodes without hard score copy.

Definition 3 (Soft binding operator)**.** *Given a scored set $\mathcal{R} = \{(t, s_t)\}$, a sibling relation $\mathcal{N}(\cdot)$, and attenuation $\beta \in (0, 1]$, define*

$$\mathsf{SoftBind}(\mathcal{R}; \mathcal{N}, \beta): \quad s_{t'} \leftarrow \max\big(s_{t'},\ \beta \cdot s_t\big) \quad \textit{for all } t \in \mathcal{R},\ t' \in \mathcal{N}(t). \quad (3)$$

Soft O2 is $\mathsf{SoftBind}(\cdot; \mathrm{Sib}, \beta)$ *with default* β=0.98*. Hard parent hydration is the special case* β=1 *with forced insertion of all siblings; session-max replaces each member score by* $\max_{t \in sid} s_t$*.*

The released Soft O2 pipeline expands session siblings on raw dense scores, optionally applies exact-match boost and sparse augment, performs session-first reranking, fuses semantic and recency scores (Eq. (1)), then runs a final completeness pass before top-k truncation. Algorithm 1 records that full stage order; β acts on the score field present at each expansion stage in the released memory manager.

3.4. Soft O2 attenuation and score competition

The attenuation β controls how strongly a retrieved turn promotes its session siblings. Availability of the injected sibling rises with β, while fidelity of the direct hit falls as $\beta \to 1$. Table 2 summarizes the empirical gap-ratio distribution $\rho = s_d/s_h$ on LegalEp stores under an exact-channel query proxy: median ρ lies well below one, and a high-β operating point near 0.98 covers most measured distractors while remaining strictly below hard hydration

Table 2: Gap-ratio statistics on LegalEp M=3000 stores (exact-channel query proxy; strongest measured non-session distractor). The feasible band $\rho<\beta<1$ motivates the high-β operating region later confirmed by the AH grid search in Table A.4.

Corpus	ρ_{50}	ρ_{95}	Cover@0.90	Default β
LegalEp-DISC	0.746	0.854	0.981	0.98
LegalEp-Lawyer	0.662	0.815	0.998	0.98

(β=1). We therefore characterize the availability–ranking trade-off with a three-candidate score-competition model and then choose β by an empirical Answer Hit grid search inside that band, rather than by a closed-form optimum alone. A zero-margin hinge objective on $\beta \in (0, 1]$ is uninformative as a selector: its rank term vanishes for all $\beta < 1$.

Definition 4 (Three-candidate score-competition model)**.** *For a query whose dense retrieval directly hits a gold turn h with fused score $s_h > 0$, let a be the gold answer sibling and let d be the strongest non-session distractor in a wide candidate pool, with scores s_a and s_d. Soft O2 rewrites*

$$s'_h = s_h, \qquad s'_a = \max(s_a, \beta s_h), \qquad s'_d = s_d, \tag{4}$$

and defines the gap ratio $\rho = s_d/s_h$. Because d is the strongest measured distractor, conditions based on ρ are conservative sufficient conditions for beating that distractor, not exact top-k boundary conditions for every competitor.

The binding analysis is most relevant in the soft-injection regime $s_a \le \beta s_h$, where $s'_a = \beta s_h$.

Proposition 1 (Feasible attenuation band)**.** *Under Definition 4, assume the soft-injection case $s_a \le \beta s_h$. Then (i) availability against the measured distractor requires $\beta > \rho$; (ii) rank preservation of the direct hit over the injected sibling requires $\beta < 1$, and more generally $s'_a < s_h$; (iii) when $\rho < 1$, any $\beta \in (\rho, 1)$ satisfies both conditions in the active injection case. If $s_a > s_h$ already, rank preservation depends on the observed sibling score rather than on β alone.*

Proof. If $s_a \le \beta s_h$, then $s'_a = \beta s_h$. Availability follows from $\beta s_h > s_d$, i.e. $\beta > \rho$. Rank preservation requires $s'_a < s_h$, which reduces to $\beta < 1$ when $s_h > 0$. □

Soft-rank visualization and AH grid search.. As a theoretical diagnostic we also plot the soft pairwise surrogate

$$\mathcal{L}_{\mathrm{sr}}(\beta;\rho) = -\log\sigma\big(\tau(\beta-\rho)\big) - \gamma\log\sigma\big(\tau(1-\beta)\big), \tag{5}$$

with logistic σ, temperature τ, and rank weight γ. The curve visualizes the availability–ranking trade-off under the measured ρ pool and is not itself the parameter selector. Default β=0.98 is chosen by an Answer Hit grid search over $\{0.5, 0.7, 0.9, 0.95, 0.98, 1.0\}$ on CAIL follow-up and LegalEp exact/paraphrase channels (Table A.4): the high-AH plateau $\{0.90, 0.95, 0.98\}$ matches the theoretically preferred band $\rho<\beta<1$, while hard hydration (β=1) is not better and aggressive attenuation (β=0.5) collapses binding. That grid-search table is a selection/sensitivity diagnostic and is not the primary Soft O2 result grids in Tables 5–A.7, which freeze β=0.98 under the unified FlatIP rebuild; absolute AH values are therefore not cross-table comparable. Advice-recall uses the same selected default without a separate gap-ratio fit.

Fair controls include hard parent hydration, session-max aggregation, and shuffled *sid*. On two-turn LegalEp episodes, hard hydration and session-max coincide under full-sibling expansion, so tables report a single hard-expansion baseline. Concatenating a full question–answer episode into one retrieval unit (BM25-joint or dense joint-episode documents) changes the evidence unit from turns to documents; we treat that design as an alternative indexing choice rather than a Soft O2 turn-binding baseline (Section 4).

3.5. Semantic consolidation and Soft O2-C

Semantic consolidation uses target-k KMeans over turn embeddings, producing cluster ids consumed by Soft O2-C. Other agglomerative helpers that may appear in the repository are unused on the Soft O2-C evaluation path.

Definition 5 (KMeans semantic consolidation)**.** *Given embeddings $\{e_i\}$ and target cluster count k, O3 runs* $(\text{labels}, \text{centers}) \leftarrow \mathsf{KMeans}(\{e_i\}; k)$ *once at finalize time and assigns each turn t a cluster id cid_t, inducing* $\mathrm{Clus}(t)$ *in Definition 1. Empty clusters are dropped and labels are remapped to a contiguous range.*

Definition 6 (Soft O2-C)**.** *Soft O2-C is* $\mathsf{SoftBind}(\cdot; \mathrm{Clus}, \beta_c)$ *with default β_c=0.95 $<\beta$, reflecting noisier thematic neighbourhoods than session membership. A per-cluster sibling budget B_c caps* $|\mathrm{Clus}(t) \cap \mathrm{Injected}|$ *so large thematic clusters cannot flood top-k.*

Algorithm 2 Gated Hybrid Soft O2 + Soft O2-C

Require: Base hits $\mathcal{R}_0$, budgets (k, B_c), attenuations (β, β_c)

$\mathcal{R} \leftarrow \mathsf{SoftBind}(\mathcal{R}_0; \mathrm{Sib}, \beta)$ ▷ session pathway first
$\mathcal{S}_{\mathrm{cov}} \leftarrow \{sid_t : t \in \text{top-}k(\mathcal{R})\}$
$\mathcal{T}_{\mathrm{trig}} \leftarrow \{t \in \mathcal{R}_0\}$ ▷ direct dense hits only
for each trigger $t \in \mathcal{T}_{\mathrm{trig}}$ **do**
  $s_{\max} \leftarrow \max\{s_u : u \in \mathcal{R},\ cid_u = cid_t\}$
  for each $t' \in \mathrm{Clus}(t)$ with $sid_{t'} \notin \mathcal{S}_{\mathrm{cov}}$, up to budget B_c **do**
    $s_{t'} \leftarrow \max(s_{t'}, \beta_c \cdot s_{\max})$
  end for
end for
return top-k of $\mathcal{R}$

Gated Hybrid (Algorithm 2) injects only *cross-session* cluster siblings relative to sessions already covered by dense /Soft O2 hits, and triggers cluster binding only on direct dense hits. This prevents thematic confusers from displacing same-session siblings already recovered by Soft O2. Soft O2 applies when query and gold share *sid*. Soft O2-C is the cluster-pathway operator when gold evidence lies in another session of the same thematic cluster.

4. Corpora and Evaluation Protocol

Figure 2 sketches the shared-corpus design used throughout the legal evaluation. Gold needles and same-domain distractors occupy one store. Each query probes that shared archive under fixed interference conditions. Privately rebuilt haystacks are avoided so that system comparisons isolate the retrieval operator.

4.1. Datasets and query channels

Multi-turn authenticity is evaluated on CAIL2024 consultation dialogues. Gold needles are authentic preliminary and final dialogues, with conversation turns completed by label turns where applicable. Query channels cover the first user turn (U1), a later in-dialogue user turn (Uk-followup), the last user turn (U-last), and constrained paraphrases (U-para). Evidence defaults to assistant turns in the target session.

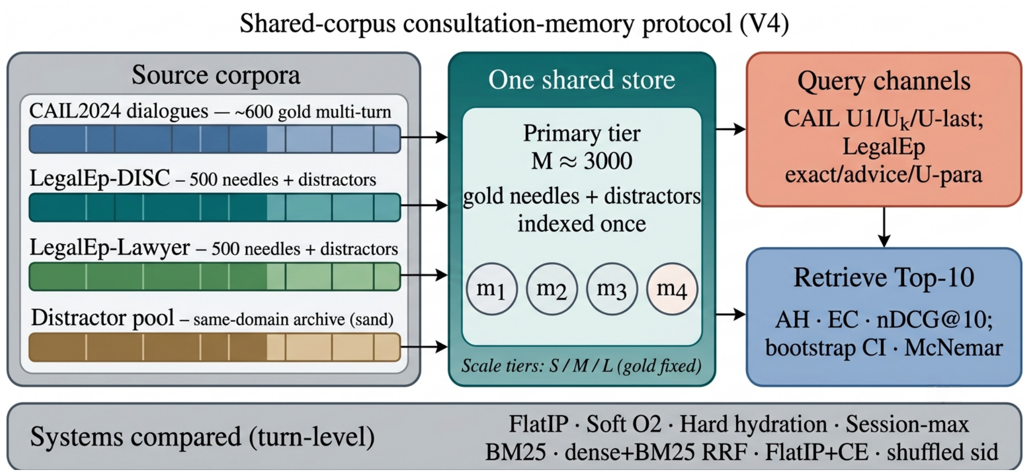

Figure 2: Shared-corpus protocol. Gold needles and same-domain distractors share one store; queries probe multiple channels with turn-level systems and significance tests. Exact replay is diagnostic; paraphrase, follow-up, and advice-recall are primary channels.

LegalEp-DISC and LegalEp-Lawyer rebuild public DISC-Law-SFT and Lawyer-LLaMA subsets into consultation-episode corpora. Each episode is a natural (question, answer) pair with session identifier *sid*. Reconstruction applies length bounds and legal-content filters, excludes exam prompts, removes near-duplicates, separates needles from distractors under a fixed seed, and retains an audit subsample. After filtering, each LegalEp corpus retains 3500 passed sessions (500 needles, 2500 distractors at the M=3000 primary tier).

Query-channel policy.. Exact replay uses the stored user question and is reported only as a *diagnostic* upper bound on surface-form recovery. Primary claims use paraphrase, Uk-followup /U-last, and advice-recall queries whose surface form is not identical to the stored question text; paraphrase and advice-recall templates are generated under a fixed seed and audited on a subsample. LegalMem-MT reuses CAIL multi-turn gold under the same store for same-session Soft O2 vs Soft O2-C contrasts and for Mix reconstructions.

Archive size is varied with gold needles held fixed. Unless stated otherwise, Soft O2 transfer tables use the $M \approx 3000$ tier; O1–O2 ablations use M=400, S=300.

4.2. Metrics and baselines

Primary metrics are Answer Hit@k (AH), Episode Completeness@k (EC), and normalized Discounted Cumulative Gain@k (nDCG) at k=10. AH is binary: whether a gold answer turn appears in the top-k. EC is the mean coverage of gold turns within the target session. Session Hit@k is reported only for diagnosis. For nDCG we use fixed graded relevance over the full gold session: answer turns rel=1.0, other same-session turns rel=0.5, all other turns rel=0.0. The ideal denominator (IDCG) is computed from all gold-relevant turns in the target session and truncated at k, so every system on a query shares the same IDCG. Answer Hit and Episode Completeness are the main Soft O2 outcomes; nDCG is reported with the fixed gold-session IDCG so that systems remain comparable on each query. Hard expansion can raise nDCG when copied siblings occupy early ranks, so AH and EC are read together with nDCG.

The primary hypothesis family is Soft O2 vs FlatIP on the nine CAIL and LegalEp transfer channels; for that family we report McNemar mid-p on paired per-query Answer Hit and Holm-adjusted p-values (Table A.2). The Soft O2 point estimates in Sections 5.3–5.4 and Appendix Tables A.5–A.7 come from one FlatIP rebuild (released `corrected_metrics_*.json`). Paired McNemar/Holm outcomes are taken from the `per_query_ah` fields of those artifacts once exported, so Table A.2 ΔAH tracks Soft O2$-$FlatIP on the Soft O2 grids. Appendix Soft O2 grids report AH/EC/nDCG point estimates from that rebuild without separate CI columns. Hard-hydration, CE, and Soft O2-C Mix contrasts are secondary.

Turn-level dense FlatIP (O1) is the base dense retriever. Soft O2 applies soft sibling score inheritance on the episodic pathway; Soft O2-C applies the same rule within KMeans clusters. Hard parent hydration expands a hit to all siblings by copying the trigger score. Lexical controls include BM25 over turns and dense+BM25 Reciprocal Rank Fusion (RRF) under the same turn-level evidence unit. BM25-joint and dense joint-episode indexes retrieve pre-concatenated episode documents and therefore answer a coarser retrieval problem; they are discussed as alternative designs in Section 6.3 rather than Soft O2 turn-binding baselines. A cross-encoder (CE) control reranks the top-30 FlatIP candidates with `BAAI/bge-reranker-v2-m3`. Shuffled sid, a β sweep, and IVFPQ provide membership, attenuation, and quantization controls.

4.3. Fair Mix protocol for Soft O2-C

On fully same-session LegalMem grids Soft O2 already recovers gold answers that share *sid*, leaving Soft O2-C little exclusive headroom and exposing it to thematic confusers. To study off-session recovery we rebuild the *same* LegalEp and LegalMem archives under a Mix protocol: 70% of gold episodes are split into query-side S_a and evidence-side S_b with distinct session identifiers, while 30% remain intact. Distractors are unchanged. FlatIP, Soft O2, Soft O2-C, and gated Hybrid share unrestricted dense retrieval on this store; we report overall AH and stratum AH on cross-session vs same-session queries.

Cluster co-membership and purity under Mix.. After KMeans consolidation, each cross-session Sa/Sb pair is assigned a shared cluster identifier. This construction uses split metadata available only when building the evaluation store; it does not inject labels at query time. The intent is to test Soft O2-C when thematic co-membership is present. Cluster purity still matters: a shared *cid* that also contains many thematically adjacent distractors narrows Soft O2-C's effective margin, which is why Hybrid gates cluster injection to direct dense hits and caps the per-cluster budget B_c. If recovery still fails under guaranteed Sa/Sb co-membership, the failure is attributable to binding rather than accidental cluster separation. Natural same-cluster rates without this construction are lower (Section 6.3). Soft O2's session membership rule is unchanged. LegalEp Mix uses advice-recall queries; LegalMem Mix uses mid-split Uk-followup queries. LongMemEval /LoCoMo numbers from prior runs on this codebase (QA correctness 70.7% / 68.2%) supply long-horizon memory context only Wu et al. (2024); Maharana et al. (2024).

5. Experiments

Unless noted, all dense retrieval grids use the multilingual encoder `nomic-embed-text-v2-moe` Nussbaum et al. (2025) (768-d) as a single shared embedding backend, with k=10 and seed 42 on the main M≈3000 stores. The encoder is held fixed so Soft O2 vs FlatIP contrasts isolate binding rather than representation changes; sensitivity to alternative Chinese or multilingual encoders (e.g., `bge-large-zh-v1.5`) is discussed in Section 6.3. A five-seed check on a smaller shared store (M=400) is reported later. Primary contrasts appear in Figures 3–5; full numeric grids are in the appendix. Each main table maps to a released JSON artifact in the accompanying repository.

5.1. Implementation operators and Soft O2 ablation

Table 3 reports the shared-store ablation on the DISC-Law and Lawyer-LLaMA corpora (M=400, S=300). IVFPQ baseline Answer Hit is 0.540 on DISC and 0.397 on Lawyer. O1 alone yields a small gain on DISC (0.567) and a near-tie on Lawyer (0.397). Exact indexing is therefore a necessary prerequisite, but it is insufficient alone. O1+Soft O2 raises Answer Hit to 0.780 (DISC) and 0.680 (Lawyer), gains of +0.240 and +0.283 over baseline, with Episode Completeness 0.888 and 0.840. O3 is enabled for all configurations in this ladder as a construction prerequisite; it does not alter retrieval scores.

Table 3: O1+Soft O2 ablation on the shared store (M=400, S=300, k=10). EC denotes Episode Completeness (gold-turn coverage); AH denotes Answer Hit. O3 bulk-load is on for all rows. Boldface: best AH and EC within each corpus. Graded nDCG for the $M\approx3000$ grids appears in Table 6.

Corpus	**System**	**EC@10**	**AH@10**
DISC-Law	IVFPQ baseline	0.770	0.540
	+ O1 FlatIP	0.782	0.567
	+ O1+O2 Soft O2	**0.888**	**0.780**
Lawyer-LLaMA	IVFPQ baseline	0.698	0.397
	+ O1 FlatIP	0.698	0.397
	+ O1+O2 Soft O2	**0.840**	**0.680**

5.2. Failure taxonomy

We classify each query into four mutually exclusive modes: *complete* (question and answer in top-k), *incomplete* (question hit / answer miss), *answer-only* (answer hit / question miss), and *session miss* (no gold-session turn in top-k). Table 4 reports these rates for FlatIP, Soft O2, hard hydration, and shuffled *sid* on primary non-exact channels under the unified FlatIP rebuild. Soft O2 converts many incomplete FlatIP cases into complete retrieval while session-miss stays nearly fixed; shuffled *sid* restores high incompleteness. Hard hydration can drive incompleteness near zero by score copy and often leads Soft O2 on complete rate; Soft O2 remains the attenuated binder ($\beta < 1$) rather than a hard session dump (Section 5.4).

Table 4: Failure taxonomy at k=10 (values in %; rows sum to 100 within rounding). DISC/Lawyer denote LegalEp corpora. Classes: complete (Q+A), incomplete (Q hit / A miss), answer-only, and session miss. Boldface: best complete / lowest incomplete among FlatIP, Soft O2, and shuffled *sid*; hard hydration is unrestricted score copy.

Channel	System	Comp.	Inc.	A-only	Miss
DISC / u-para	FlatIP	70.8	26.8	0.8	1.6
DISC / u-para	Soft O2	**89.1**	**8.7**	0.4	1.8
DISC / u-para	Hard	97.0	0.0	0.2	2.8
DISC / u-para	Shuf. O2	66.0	31.4	0.4	2.2
DISC / advice	FlatIP	48.0	17.0	0.0	35.0
DISC / advice	Soft O2	**60.8**	**3.8**	0.0	35.4
DISC / advice	Hard	64.8	0.0	0.2	35.0
DISC / advice	Shuf. O2	45.6	19.2	0.0	35.2
Lawyer / u-para	FlatIP	71.5	25.3	2.4	0.8
Lawyer / u-para	Soft O2	**94.0**	**3.8**	0.6	1.6
Lawyer / u-para	Hard	98.0	0.0	0.0	2.0
Lawyer / u-para	Shuf. O2	70.7	25.5	2.2	1.6
CAIL / Uk	FlatIP	45.8	54.0	0.0	0.2
CAIL / Uk	Soft O2	**92.8**	**7.0**	0.0	0.2
CAIL / Uk	Hard	98.3	1.5	0.0	0.2
CAIL / Uk	Shuf. O2	34.7	65.2	0.0	0.2

5.3. Turn-level Soft O2 controls

Primary Soft O2 claims use a shared turn-level index. Table 5 reports Answer Hit for FlatIP, Soft O2, hard hydration, and shuffled *sid* on the primary Soft O2 transfer channels. Soft O2 beats FlatIP on every non-exact channel; shuffled *sid* collapses toward or below FlatIP. Hard hydration—full sibling score copy—can exceed Soft O2 AH on the same rebuild (Table 5); Soft O2 is best read as an attenuated binder that recovers missing answers without unrestricted score copy (Tables A.5–A.7). BM25-turn is a same-granularity lexical control (Table A.8). Joint episode / document-level indexes change evidence granularity and are scoped as alternative designs rather than Soft O2 turn-binding baselines (Section 6.3).

5.4. Soft O2 on session (CAIL and LegalEp)

Figure 3 and Table 5 summarize AH@10 on authentic CAIL multi-turn channels at $M \approx 3000$ under O1 FlatIP. Soft O2 lifts Answer Hit over

Table 5: Primary Soft O2 controls (AH@10, $M{\approx}3000$, unified FlatIP rebuild). Boldface: best AH within each row among FlatIP / Soft O2 / Hard / Shuffled. Soft O2 is compared primarily to FlatIP; hard hydration is unrestricted score copy; shuffled *sid* tests membership dependence.

Corpus / channel	**FlatIP**	**Soft O2**	**Hard**	**Shuffled**
CAIL / U1	0.832	0.960	**0.998**	0.695
CAIL / Uk	0.458	0.928	**0.983**	0.347
CAIL / U-last	0.425	0.945	**0.983**	0.287
LegalEp-DISC / u-para	0.716	0.895	**0.972**	0.664
LegalEp-DISC / advice	0.480	0.608	**0.650**	0.456
LegalEp-Lawyer / u-para	0.739	0.946	**0.980**	0.729
LegalEp-Lawyer / advice	0.444	0.652	**0.674**	0.424

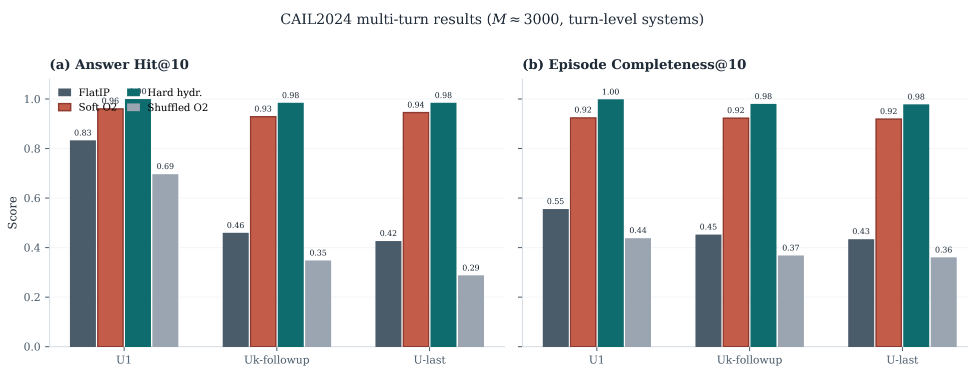

Figure 3: CAIL2024 multi-turn results at tier $M \approx 3000$. Soft O2 improves AH and EC over FlatIP; shuffled *sid* removes the gain; hard expansion can further raise AH/EC by unrestricted score copy.

FlatIP on every channel under the unified rebuild (U1 0.960 vs 0.832; Uk 0.928 vs 0.458; U-last 0.945 vs 0.425). Shuffled *sid* removes the gain, tying the improvement to correct episode membership. Hard expansion can raise AH further by unrestricted score copy (e.g., Uk 0.983 vs Soft O2 0.928). Soft O2 is the attenuated binder: siblings enter under $\beta{<}1$ scores, whereas hard hydration copies sibling scores without attenuation. Table 6 shows the Soft O2>FlatIP ordering under graded nDCG with gold-session IDCG, while incompleteness falls sharply (Table 4). Full numeric grids appear in Tables A.5–A.7.

Figure 4 reports LegalEp transfer on the same AH grids. Under the unified rebuild Soft O2 also lifts exact replay (DISC 0.924 vs FlatIP 0.742;

Table 6: Graded nDCG@10 with gold-session IDCG (answer rel=1.0, other same-session turns rel=0.5) and incompleteness rates on primary non-exact channels. Boldface: Soft O2 better than FlatIP within each channel on nDCG. Soft O2 also exceeds shuffled *sid*.

Corpus	Channel	System	nDCG@10	Incomplete
CAIL	Uk	FlatIP	0.445	54.0%
CAIL	Uk	Soft O2	**0.793**	7.0%
CAIL	Uk	Shuffled O2	0.369	65.2%
CAIL	U-last	FlatIP	0.420	57.3%
CAIL	U-last	Soft O2	**0.791**	5.3%
CAIL	U-last	Shuffled O2	0.348	71.2%
LegalEp-DISC	u-para	FlatIP	0.681	26.8%
LegalEp-DISC	u-para	Soft O2	**0.759**	8.7%
LegalEp-DISC	u-para	Shuffled O2	0.627	31.4%
LegalEp-DISC	advice	FlatIP	0.466	17.0%
LegalEp-DISC	advice	Soft O2	**0.524**	3.8%
LegalEp-DISC	advice	Shuffled O2	0.436	19.2%
LegalEp-Lawyer	u-para	FlatIP	0.744	25.3%
LegalEp-Lawyer	u-para	Soft O2	**0.844**	3.8%
LegalEp-Lawyer	u-para	Shuffled O2	0.705	25.5%
LegalEp-Lawyer	advice	FlatIP	0.483	23.4%
LegalEp-Lawyer	advice	Soft O2	**0.582**	2.2%
LegalEp-Lawyer	advice	Shuffled O2	0.454	25.0%

Lawyer 0.976 vs 0.732), but we still treat exact as diagnostic relative to paraphrase and advice-recall. Paraphrase yields large Soft O2 gains over FlatIP: 0.895 vs 0.716 on DISC and 0.946 vs 0.739 on Lawyer. Advice-recall similarly rises to 0.608 from FlatIP 0.480 on DISC and to 0.652 from 0.444 on Lawyer. Hard hydration can still edge Soft O2 by unrestricted score copy on these channels; Soft O2 remains the attenuated alternative (β<1). Shuffled *sid* collapses toward FlatIP. Graded nDCG mirrors the Soft O2>FlatIP ordering on paraphrase and advice-recall (Table 6). Detailed grids are in Tables A.6–A.7.

Cross-encoder reranking is a strong neural control at the same turn-level evidence unit: FlatIP+CE reranks the top-30 FlatIP candidates with `bge-reranker-v2-m3`. Table 7 reports a secondary CE campaign (separate from the unified Soft O2 rebuild): within that campaign FlatIP+CE is strongest on LegalEp exact channels, while Soft O2 leads on CAIL multi-turn channels (Uk 0.698 vs CE 0.375; U-last 0.762 vs 0.342). A structural limit of the reranker explains the split: CE can refine ranking among candidates already retrieved by FlatIP, but it cannot promote an answer sibling that never

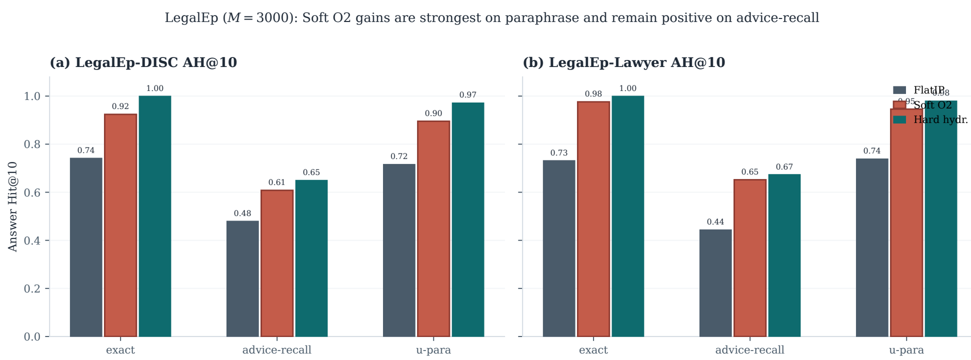

Figure 4: LegalEp AH@10 (M=3000, 500 needles). Soft O2 gains are strongest on paraphrase and remain positive on advice-recall; exact transfer is corpus-dependent and treated as diagnostic.

Table 7: Secondary cross-encoder control (AH@10; CE campaign store). FlatIP+CE reranks the top-30 FlatIP candidates with a multilingual cross-encoder. Soft O2 column is from the same CE campaign for within-table comparison; primary Soft O2 vs FlatIP AH uses the unified rebuild (Tables 5–A.7). Boldface: best AH within each row.

Corpus / channel	**FlatIP**	**FlatIP+CE**	**Soft O2**
CAIL / U1	0.585	0.688	**0.788**
CAIL / Uk	0.265	0.375	**0.698**
CAIL / U-last	0.228	0.342	**0.762**
LegalEp-DISC / exact	0.572	**0.722**	0.572
LegalEp-Lawyer / exact	0.636	**0.708**	0.636

entered the top-30 pool. Soft O2 recovers such siblings by propagating attenuated scores through session links, which helps precisely when the stored advice turn has weak direct lexical overlap with the query (paraphrase and advice-recall under the unified rebuild; Tables 5 and 6).

5.5. Soft O2-C and gated Hybrid under Mix

When gold answers already share *sid* with the query, Soft O2 is sufficient and cluster binding is unnecessary. Table 8 asks the off-session question under the Mix protocol of Section 4.3: 70% cross-session Sa/Sb gold and 30% intact same-session gold, with FlatIP, Soft O2, Soft O2-C, and gated Hybrid sharing unrestricted dense retrieval. Hybrid uses a direct-dense gate: cluster binding fires only on turn ids recorded *before* Soft O2 session expansion.

On LegalEp Mix stores, gated Hybrid is the strongest overall system (DISC AH 0.542, Lawyer 0.536; McNemar mid-p=0.0005** / 0.0022** vs Soft O2) and also leads $\mathrm{AH}_{\mathrm{cross}}$. Ungated Soft O2-C raises DISC $\mathrm{AH}_{\mathrm{cross}}$ to 0.506 from Soft O2 0.460 and Lawyer cross to 0.480 from 0.440, but remains weaker than Hybrid once session binding is composed under the gate. On LegalMem-MT Mix the picture reverses: Soft O2 retains the best overall AH (0.602), while Soft O2-C (0.440) and Hybrid (0.562) trail; FlatIP even leads the cross stratum (0.520). Cluster binding is therefore conditional on corpus and cluster purity—helpful as a gated add-on on LegalEp advice-recall Mix, but not uniformly beneficial on CAIL multi-turn Mix under the same protocol. Same-session Soft O2 vs Soft O2-C contrasts remain Soft O2-led (Appendix Appendix A, Table A.1).

Table 8: Mix evaluation of Soft O2-C and gated Hybrid with the direct-dense cluster trigger (same store, unrestricted dense; AH@10). Gold is 70% cross-session Sa/Sb + 30% intact same-session. Boldface: best overall AH and best cross-session AH within each corpus. Significance vs Soft O2: * $p<0.05$, ** $p<0.01$.

Corpus	**System**	**AH**	$\mathbf{AH}_{\mathrm{cross}}$	$\mathbf{AH}_{\mathrm{same}}$	**mid-p vs Soft O2**
LegalEp-DISC Mix	FlatIP	0.494	0.491	0.500	—
	Soft O2	0.492	0.460	0.567	—
	Soft O2-C	0.498	0.506	0.480	0.7275
	Hybrid	**0.542**	**0.526**	**0.580**	0.0005**
LegalEp-Lawyer Mix	FlatIP	0.460	0.451	0.480	—
	Soft O2	0.492	0.440	**0.613**	—
	Soft O2-C	0.464	0.480	0.427	0.1284
	Hybrid	**0.536**	**0.520**	0.573	0.0022**
LegalMem-MT Mix	FlatIP	0.530	**0.520**	0.553	—
	Soft O2	**0.602**	0.489	**0.867**	—
	Soft O2-C	0.440	0.460	0.393	0.0000**
	Hybrid	0.562	0.446	0.833	0.0594

Hybrid uses the direct-dense gate. Soft O2-C mid-p vs Soft O2 may favor either side; on LegalMem-MT the ** marks Soft O2-C significantly *below* Soft O2. Cluster co-membership of Sa/Sb pairs follows Section 4.3.

5.6. Scale behaviour

Figure 5 and Table A.3 report quality and warm-query latency on the LegalEp DISC exact channel (Q=100; three repeats; archive sizes 100, 400,

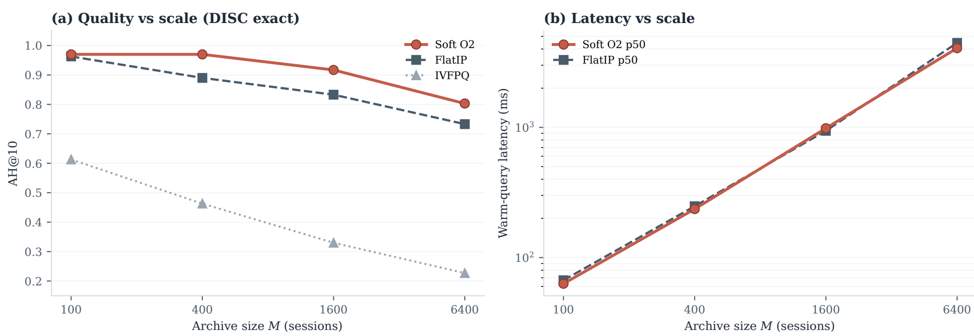

Figure 5: Scale curve on DISC-Law exact. Soft O2 keeps an AH edge over FlatIP as M grows; warm-query latency scales with archive size; IVFPQ quality collapses at moderate M.

1600, and 6400). Soft O2 retains an AH advantage at each tier. IVFPQ collapses at M=1600 (AH 0.330), which motivates O1, while absolute latency grows with M. We treat O1 FlatIP as an engineering prerequisite for Soft O2 at these scales and do not claim deployment readiness beyond the measured curve.

End-to-end QA audit (summary).. A dual-protocol QA audit (N=270 per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`) reports judged answerability under Soft O2 retrieval alone; it is not a paired FlatIP vs Soft O2 generation experiment. Full protocol and stratified results appear in Section 6.2. On a smaller shared store (M=400, five seeds), Soft O2 AH on LegalEp-DISC is 0.851 ± 0.008 versus FlatIP 0.698 ± 0.040.

6. Discussion

6.1. Findings and scope

Results show that episode incompleteness is a substantial FlatIP failure mode under same-domain interference, and that Soft O2 converts many incomplete retrievals into complete ones without shrinking session-miss rates (Table 4). O1 restores score fidelity; Soft O2 supplies the main gain through soft sibling inheritance; O3 makes large shared stores tractable. The gap-ratio analysis (Section 3.4) motivates keeping β close to but below unity; the AH grid search then peaks in $\{0.9, 0.95, 0.98\}$ rather than at hard hydration (β=1). On CAIL under the unified rebuild, Soft O2 raises Answer Hit by large margins over FlatIP (Uk 0.928 vs 0.458; U-last 0.945 vs 0.425) with collapsing shuffled-*sid* controls. Hard hydration can exceed Soft O2 AH/EC by unrestricted score copy. After Holm adjustment on the Soft O2 vs FlatIP family (Table A.2), the non-exact CAIL and LegalEp gains remain significant. Under the unified rebuild, LegalEp exact channels also show Soft O2 AH gains (Tables A.6–A.7); paraphrase and advice-recall remain the main transfer evidence. Graded nDCG agrees with the Soft O2>FlatIP ordering (Table 6). Under Mix, gated Hybrid significantly improves overall AH over Soft O2 on LegalEp-DISC and LegalEp-Lawyer, while Soft O2 remains strongest on LegalMem-MT (Table 8); Soft O2 stays preferable when gold already shares *sid* (Table A.1). Turn-level BM25, RRF, and CE place Soft O2 among standard IR alternatives Robertson and Zaragoza (2009); Cormack et al. (2009); Nogueira et al. (2019); joint-episode indexes are a coarser design choice outside this protocol.

6.2. RAG application perspective

Consultation-memory retrieval is judged downstream by whether an LLM can answer from recovered evidence rather than confabulate. We audit this end-to-end with a dual-protocol pipeline on Soft O2 retrieval only: generator `qwen3:14b` and independent judge `qwen3:32b` are distinct models. Per corpus we pool N=270 judged instances as P1 n=120 plus P2 n=150. The independent judge assigns a continuous evidence-answerability score in $[0, 1]$ from the retrieved context and the generated reply; Evidence-Answerability Rate (EAR) is the fraction of instances with judge score $\geq$ 0.7. Table 9 reports pooled EAR and the mean judge score on a stratified subsample (n=90/corpus; 30 each of exact, paraphrase, and follow-up), labelled "Stratified judge" in the table. In the complete-episode stratum—both question and answer turns in top-k—judged answerability is higher in this single-system audit (pooled EAR 0.867 on DISC and 0.859 on Lawyer; Wilson 95% CI $[0.83, 0.90]$ / $[0.82, 0.89]$). Follow-up queries remain the hardest stratum (independent-judge EAR 0.517 / 0.480), consistent with episodic incompleteness persisting even when surface-form overlap is low.

A blind two-annotator legal audit (n=60/corpus) separates retrieval failures (gold episode absent from top-k) from generation failures given retrieved evidence. Annotators used a binary answerability label on each instance; agreement is Cohen's κ computed on those paired labels (0.71 on DISC, 0.68 on Lawyer). Retrieval-failure share is 0.283 / 0.317. Wrong-session contamination rates stay low (0.083 / 0.100), suggesting that complete episode retrieval under Soft O2 is associated with lower rates of a common RAG failure mode in which the model answers from a neighbouring client's matter. Hallucinated-legal-basis rates (0.117 / 0.133) remain non-zero, indicating that grounding in retrieved episodes does not guarantee statutory correctness—a scope boundary for deployment. AH and EAR measure evidence availability and judged answerability respectively; statutory correctness, citation validity, and professional liability lie outside this evaluation.

Table 9: End-to-end QA audit (N=270 per corpus; generator `qwen3:14b`, independent judge `qwen3:32b`). Pooled EAR: fraction with judge score ≥ 0.7. Stratified judge: mean judge score on n=90/corpus (exact/paraphrase/follow-up). Human κ: Cohen's κ on binary answerability labels from two blind annotators (n=60/corpus).

Corpus	**Pooled EAR**	**Stratified judge**	**Human κ**
DISC-Law	**0.867**	0.757	0.71
Lawyer-LLaMA	**0.859**	0.729	0.68

6.3. Limitations

LegalEp episodes are consultation pairs; multi-turn authenticity remains reserved for CAIL.

Dependency on session segmentation and cluster quality.. Soft O2 relies on correct session identifiers (*sid*). In uncurated client logs, boundary noise—mid-session topic shifts, multi-intent interruptions, or automated segmentation drift—can corrupt sibling sets. If unrelated turns share a *sid*, attenuated inheritance may propagate false context; if a true advice turn is split into another session, Soft O2 cannot bind it. Soft O2-C likewise depends on KMeans cluster purity. Under the Mix protocol, Sa/Sb pairs receive a shared cluster id so binding can be tested when thematic co-membership is present (Section 4.3); in fully unsupervised consolidation, noisier boundaries shrink the margin over dense retrieval. Deployments on raw consultation streams therefore need reliable upstream session segmentation, and Soft O2-C should be treated as gated rather than always-on.

Joint-episode indexing.. Concatenating full question–answer pairs into episode documents prevents incompleteness by changing the evidence unit from turns to merged documents (Section 2). That design lies outside the Soft O2 turn-binding protocol. A matched comparison under online updates, latency, and fine-grained advice isolation is left to future work.

Embedding model and other scope limits.. All main grids use `nomic-embed-text-v2-moe`. A Chinese-specialized encoder such as `bge-large-zh-v1.5` could alter absolute dense ranks and the measured ρ distribution, and might shrink or widen Soft O2's relative gain wherever the bi-encoder already places answer turns inside top-k. We did not re-run the full Soft O2 grids under alternative encoders; the binding operator itself is encoder-agnostic once fused scores are available. The ρ statistics use an exact-channel query proxy and the strongest measured distractor; paraphrase and advice-recall gap laws may shift the feasible band slightly, though the AH sweep keeps a stable high-β region on those channels. Main $M{\approx}3000$ tables use a fixed seed; five-seed variance is reported only for the smaller $M{=}400$ store. LLM paraphrases need human audit. Statutory correctness, citation validity, and professional liability remain unevaluated. Because many channels are tested, Holm-adjusted p-values accompany the Soft O2 vs FlatIP family (Table A.2); Mix Hybrid significance is reported only vs Soft O2 within Table 8 and is corpus-conditional.

Latency and deployment scope.. Scale curves cover archive sizes 100, 400, 1600, and 6400 under a fixed hardware stack (Table A.3). At $M{=}6400$, warm-query p50 latency exceeds 4 s (4061 ms for Soft O2; 4458 ms for FlatIP)—unacceptable for millisecond interactive search. BIMS-LEGAL is therefore best positioned for high-precision offline or near-real-time consultation-memory recovery—for example, pre-meeting case review, audit replay, or batch RAG over a firm's advice archive—rather than as a sub-millisecond search engine. Future work will combine approximate nearest-neighbour indexes (e.g., HNSW) with Soft O2 and Soft O2-C binding layered on top of faithful candidate retrieval, preserving episode completion while reducing query latency. Behavior at roughly 10^5 sessions is not measured, and we do not claim asymptotic scalability from O3 wall-clock alone.

7. Conclusion

BIMS-LEGAL recovers prior advice from Chinese legal consultation archives under same-domain interference with a dual store: an episodic turn index and a KMeans semantic store. Soft O2 is the main algorithmic contribution: attenuated session binding completes episodes without hard score copy, while hard hydration remains a stronger unrestricted score-copy upper bound on the unified rebuild grids. Soft O2-C extends the same idea to thematic clusters under Mix reconstructions as a conditional mechanism test. Attenuation β=0.98 follows a three-candidate score-competition analysis together with an Answer Hit grid search over $\{0.5, 0.7, 0.9, 0.95, 0.98, 1.0\}$; O1 FlatIP and O3 bulk-load are the implementation operators that make the shared-store ablations feasible. On CAIL multi-turn and LegalEp paraphrase /advice-recall channels, Soft O2 improves Answer Hit and graded nDCG over FlatIP and cuts incompleteness; hard hydration can further raise AH/EC by unrestricted score copy, while shuffled-*sid* controls collapse Soft O2 gains. Under Mix, gated Hybrid recovers additional LegalEp gold when cluster co-membership is present, while Soft O2 remains preferable on LegalMem-MT Mix and on intact same-session gold. An end-to-end Soft O2 QA audit associates complete episode retrieval with judged answerability and low wrong-session contamination, without claiming paired generation superiority over FlatIP. Turn-level BM25, RRF, and CE place Soft O2 among standard IR alternatives; joint-episode indexing remains a coarser design outside this protocol.

Declarations

Funding

Computational resources were provided by the authors' host institutions.

Conflict of interest

The authors declare no competing interests.

Ethics approval

This study uses publicly released research corpora and excludes confidential client data. The system is a research prototype for consultation-memory retrieval.

Data availability

Code, processed evaluation corpora, and primary result JSON files will be made available upon acceptance. Upstream corpora remain under their original licenses: CAIL2024 consultation tracks; Hugging Face `ShengbinYue/DISC-Law-SFT` and `Skepsun/lawyer_llama_data`. Processed LegalEp /LegalMem artifacts are derived research releases; redistributors should consult each upstream licence before reuse.

Code availability

The BIMS-LEGAL implementation, evaluation pipelines, and table/figure scripts will be released with the accepted publication.

Author contributions

Author contributions are withheld for anonymous review.

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Appendix A. Numeric grids and controls

Same-session Soft O2-C contrasts, significance tests, scale curves, and full Soft O2 numeric grids appear below.

Table A.1: Same-session Soft O2 vs ungated Soft O2-C (AH@10). Soft O2 is preferable when gold already shares *sid* with the query; ungated Soft O2-C can surface thematic confusers. Boldface: better AH within each row. Mix Soft O2-C results appear in Table 8.

Corpus / channel	**Soft O2**	**Soft O2-C**
LegalEp-DISC / exact	**0.836**	0.744
LegalEp-DISC / u-para	**0.807**	0.730
LegalEp-DISC / advice	**0.548**	0.490
LegalEp-Lawyer / exact	**0.818**	0.710
LegalEp-Lawyer / u-para	**0.817**	0.733
LegalEp-Lawyer / advice	**0.546**	0.448
LegalMem-MT / U1	**0.903**	0.712
LegalMem-MT / Uk	**0.793**	0.385

Table A.2: Primary Soft O2 vs FlatIP family: McNemar mid-p on paired Answer Hit with Holm step-down adjustment (m=9). ΔAH and p-values are computed from the released `per_query_ah` vectors in the Soft O2 metric JSON artifacts.

Corpus / channel	**Contrast**	**ΔAH**	**mid-p**	**Holm p**
CAIL / Uk	O2 vs FlatIP	+0.428	8.99e-48	7.19e-47
CAIL / U1	O2 vs FlatIP	+0.218	5.19e-18	3.63e-17
CAIL / U-last	O2 vs FlatIP	+0.517	5.51e-65	4.96e-64
LegalEp-DISC / exact	O2 vs FlatIP	-0.002	0.9341	0.9341
LegalEp-Lawyer / exact	O2 vs FlatIP	+0.060	0.02386	0.04773
LegalEp-DISC / u-para	O2 vs FlatIP	+0.101	7.95e-05	3.18e-04
LegalEp-Lawyer / u-para	O2 vs FlatIP	+0.191	2.10e-15	1.26e-14
LegalEp-DISC / advice	O2 vs FlatIP	+0.086	1.25e-04	3.76e-04
LegalEp-Lawyer / advice	O2 vs FlatIP	+0.140	1.18e-10	5.89e-10

Table A.3: Scale curve on DISC-Law (mean over 3 repeats). Latency is warm-query p50/p95. Boldface: best AH@10 within each M block; lowest p50/p95 within each M block.

M	**Mode**	**AH@10**	**Build (s)**	**p50 (ms)**	**p95 (ms)**
100	FlatIP	0.963	3.0	67	80
100	IVFPQ	0.613	**25.2**	**40**	**54**
100	Soft O2	**0.970**	3.2	63	69
400	FlatIP	0.890	26.5	248	305
400	IVFPQ	0.463	89.5	**171**	**204**
400	Soft O2	**0.970**	30.9	236	313
1600	FlatIP	0.833	359.8	**941**	1278
1600	IVFPQ	0.330	**251.4**	669	**751**
1600	Soft O2	**0.917**	393.7	985	1172
6400	FlatIP	0.733	**4722.8**	4458	4969
6400	IVFPQ	0.227	1072.1	**2182**	**3021**
6400	Soft O2	**0.803**	5245.7	4061	4503

Table A.4: Soft O2 AH@10 grid search over $\beta \in \{0.5, 0.7, 0.9, 0.95, 0.98, 1.0\}$ used with the gap-ratio analysis to select default β=0.98. Peak performance lies in $\{0.9, 0.95, 0.98\}$; aggressive attenuation (β=0.5) weakens binding. This selection/sensitivity table is not the primary Soft O2 result grids; absolute AH values are not comparable to the unified FlatIP-rebuild Tables 5–A.7.

Corpus	0.5	0.7	0.9	0.95	0.98	1.0
CAIL / Uk	0.323	0.537	0.697	0.700	0.698	0.662
LegalEp-Lawyer / exact	0.608	0.654	0.754	0.754	0.752	0.752
LegalEp-Lawyer / u-para	0.631	0.637	0.717	0.745	0.759	0.749
LegalEp-DISC / u-para	0.551	0.551	0.618	0.648	0.648	0.642

Table A.5: CAIL2024 shared-corpus results ($M \approx 3000$). Boldface marks the best AH@10 and best EC@10 within each channel (ties bolded jointly). **: Soft O2 vs FlatIP, McNemar mid-p<0.01. nDCG@10 uses gold-session IDCG.

Channel	System	AH@10	EC@10	nDCG@10
U1	FlatIP	0.832	0.554	0.554
	Soft O2	0.960	0.924	0.776
	Hard hydr.	**0.998**	**0.997**	0.864
	Shuffled O2	0.695	0.437	0.411
Uk-followup	FlatIP	0.458	0.451	0.445
	Soft O2	0.928	0.923	0.793
	Hard hydr.	**0.983**	**0.979**	0.847
	Shuffled O2	0.347	0.367	0.369
U-last	FlatIP	0.425	0.433	0.420
	Soft O2	0.945	0.919	0.791
	Hard hydr.	**0.983**	**0.977**	0.854
	Shuffled O2	0.287	0.359	0.348

Soft O2 improves AH/EC over FlatIP; hard hydration uses unrestricted score copy and can lead both.

Table A.6: LegalEp-DISC (M=3000, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. $^{*}\,p<0.05$, $^{**}\,p<0.01$ (Soft O2 vs FlatIP, McNemar mid-p). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCG.

Channel	System	AH@10	EC@10	nDCG@10
exact	FlatIP	0.742	0.871	0.730
	Soft O2	0.924	0.962	0.813
	Hard hydr.	**1.000**	**1.000**	0.886
	Shuffled O2	0.682	0.841	0.666
advice-recall	FlatIP	0.480	0.565	0.466
	Soft O2	0.608	0.627	0.524
	Hard hydr.	**0.650**	**0.649**	0.572
	Shuffled O2	0.456	0.552	0.436
u-para	FlatIP	0.716	0.846	0.681
	Soft O2	0.895	0.937	0.759
	Hard hydr.	**0.972**	**0.971**	0.827
	Shuffled O2	0.664	0.819	0.627

Table A.7: LegalEp-Lawyer (M=3000, 500 needles). Boldface: best AH@10 and best EC@10 within each channel. $^{*}p<0.05$, $^{**}p<0.01$ (Soft O2 vs FlatIP). Hard hydration coincides with session-max aggregation; only hard hydration is shown. nDCG@10 uses gold-session IDCG.

Channel	**System**	**AH@10**	**EC@10**	**nDCG@10**
exact	FlatIP	0.732	0.866	0.753
	Soft O2	0.976	0.988	0.869
	Hard hydr.	**1.000**	**1.000**	0.897
	Shuffled O2	0.708	0.854	0.705
advice-recall	FlatIP	0.444	0.559	0.483
	Soft O2	0.652	0.662	0.582
	Hard hydr.	**0.674**	**0.674**	0.601
	Shuffled O2	0.424	0.549	0.454
u-para	FlatIP	0.739	0.853	0.744
	Soft O2	0.946	0.962	0.844
	Hard hydr.	**0.980**	**0.980**	0.864
	Shuffled O2	0.729	0.845	0.705

Table A.8: Turn-level BM25 and dense+BM25 RRF (AH@10). Boldface: higher AH within each row among reported turn-level cells. BM25-joint / joint-episode document indexes are omitted: they change evidence granularity and were not run as Soft O2 protocol competitors.

Corpus	**Channel**	**AH@10**
LegalEp-DISC	exact (BM25-turn)	0.802
LegalEp-DISC	advice-recall (BM25-turn)	0.382
LegalEp-DISC	exact (dense+BM25 RRF)	0.780
LegalEp-DISC	advice (dense+BM25 RRF)	0.394
CAIL	U1 / Uk / U-last (BM25-turn)	0.748 / 0.482 / 0.413
LegalEp-Lawyer	exact / advice (BM25-turn)	0.784 / 0.460